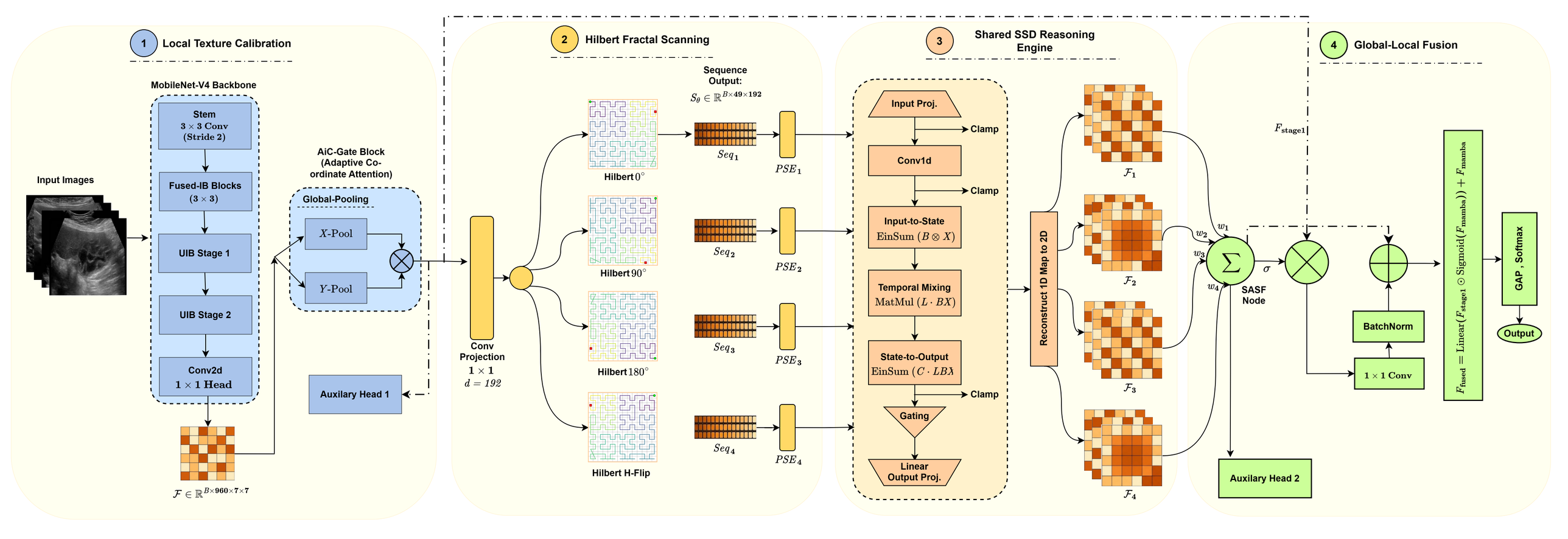

1
Local Texture Calibration
MobileNet-V4 Backbone
Input Images
Stem
3 × 3 Conv
(Stride 2)
Fused-IB Blocks
(3 × 3)
UIB Stage 1
UIB Stage 2
Conv2d
1 × 1 Head
$\mathcal{F} \in \mathbb{R}^{B\times 960\times 7\times 7}$
AiC-Gate Block
(Adaptive Co-
ordinate Attention)
Global-Pooling
X-Pool
Y-Pool
Auxilary Head 1
2
Hilbert Fractal Scanning
Conv
Projection
1 × 1
d = 192
Hilbert 0°
Hilbert 90°
Hilbert 180°
Hilbert H-Flip
Sequence
Output:
$S_\theta \in \mathbb{R}^{B\times 49\times 192}$
Seq_1
Seq_2
Seq_3
Seq_4
PSE_1
PSE_2
PSE_3
PSE_4
3
Shared SSD Reasoning
Engine
Input Proj.
Clamp
Conv1d
Clamp
Input-to-State
EinSum $(B \otimes X)$
Temporal Mixing
MatMul $(L \cdot BX)$
State-to-Output
EinSum $(C \cdot LBX)$
Clamp
Gating
Linear
Output Proj.
Reconstruct 1D Map to 2D
$\mathcal{F}_1$
$\mathcal{F}_2$
$\mathcal{F}_3$
$\mathcal{F}_4$
4
Global-Local Fusion
F_{stage1}
w_1
w_2
w_3
w_4
SASF
Node
σ
BatchNorm
1 × 1 Conv
Auxilary Head 2
$F_{\text{fused}} = \text{Linear}(F_{\text{stage1}} \odot \text{Sigmoid}(F_{\text{mamba}})) + F_{\text{mamba}}$
GAP , Softmax
Output